

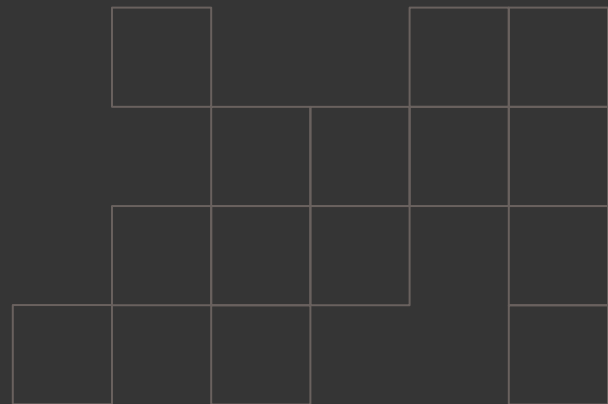
IDX Exchange - Data Science
Internship

California Home Price Prediction

A Machine Learning Approach

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Agenda



1. The problem we are solving

2. Models Explored

3. Performance Comparison

4. Performance by Price Band

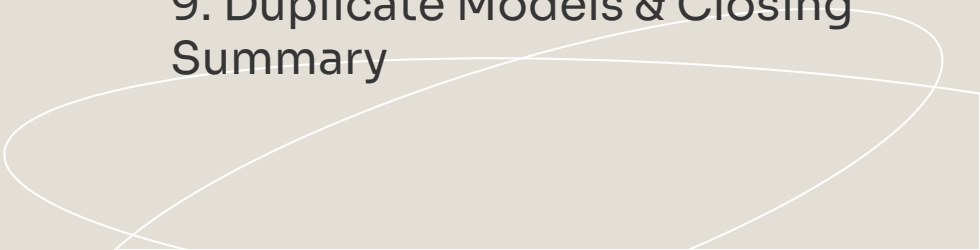
5. Stability Over Time (Lagged Test Data)

6. Iteration Timeline

7. What Worked Best

8. What Didn't Work

9. Duplicate Models & Closing Summary



The problem we are solving

California is not one housing market...

- There are dozens of distinct regional markets that are represented by a single statewide model
- Location matters more than the physical attributes of the house
- MLS data is completed by agents and lacks qualitative data
 - There are no fields for qualitative data like “recently renovated,” “ocean view,” “designed by notable architect”
- Coastal metro areas and Central Valley prices differ by multiples
 - Median home price in San Mateo County is \$2,100,000 vs \$383,250 in Kern County.
- Price distribution has a long tail representing the luxury market
 - The mean sits well above the median so optimizing for raw dollars skews the model too much on mansions and not the typical home

Models Explored

Model	Why we used it	What it contributed
Linear Regression	Fast, interpretable, price is roughly log-linear to size and location	Used as baseline model
Decision Tree	Captures non-linear trends in the data, easy to explain	Showed where a single model overfits
Random Forest	Averages many trees, keeps interactions, controls variance	First large accuracy gain
XGBoost	Sequential boosting, great for tabular data, bias/variance control	Best performing model

Performance Comparison

- Best model: Tuned XGBoost
 - Lowest MdAPE (7.76%): for 50% (half) of the properties, the predicted close price deviates from the actual close price by 7.76% or less
 - Lowest MAPE (11.69%): predicted close price deviates from the actual close price by 11.69% on average
 - Highest R^2 (0.9374): explains the most variability in close price of a home
 - Smallest MAPE-MdAPE gap (3.93 percentage points): suggests the errors in this model is less influenced by extreme outliers than other models, indicating rather consistent prediction performance overall

Model (n = 12827)	R^2	MAPE(%)	MdAPE(%)	MAPE-MdAPE Gap
Linear Regression	0.8738	17.85	12.73	5.12
Decision Tree (max_depth = 10)	0.8738	16.89	11.37	5.52
Random Forest (150 trees)	0.9255	12.68	8.20	4.48
XGBoost (default)	0.9288	12.66	8.49	4.17
XGBoost (tuned – n_estimators=1857, learning_rate=0.05, max_depth=6)	0.9374	11.69	7.76	3.93

Performance by Price Band

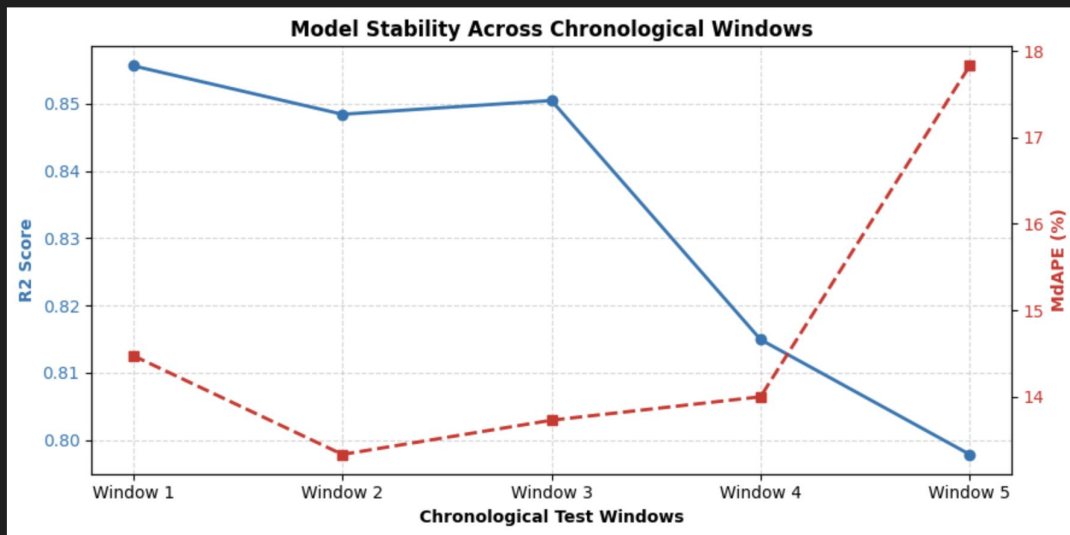
- Best model: Nate's Tuned XGBoost
- **Strongest tier:** \$500K-\$1M
 - Lowest MdAPE score of 6.06%, and a small MAPE - MdAPE gap of 2.87%
- **Low End:** Model overpredicts cheap homes which could be associated with the location-based features pulling the prediction to a typical price
 - Largest MAPE - MdAPE gap (8.08): represents outlier/unusual sales affecting the mean error
- **High End:** Model underpredicts luxury homes due to sparse data and missing features (views, land value) that quantity based metrics don't capture

Price Band	n	R ²	MAPE (%)	MdAPE (%)	MAPE - MdAPE Gap	Bias (%)
Under \$500k	1,808	0.3626	16.71	8.63	8.08	7.91
\$500k - \$1M	5,326	0.5592	8.93	6.06	2.87	1.51
\$1M - \$2M	3,945	0.3787	11.14	8.57	2.57	-0.45
+2M+	1,748	0.6950	15.97	12.80	3.17	-6.30

Stability Over Time

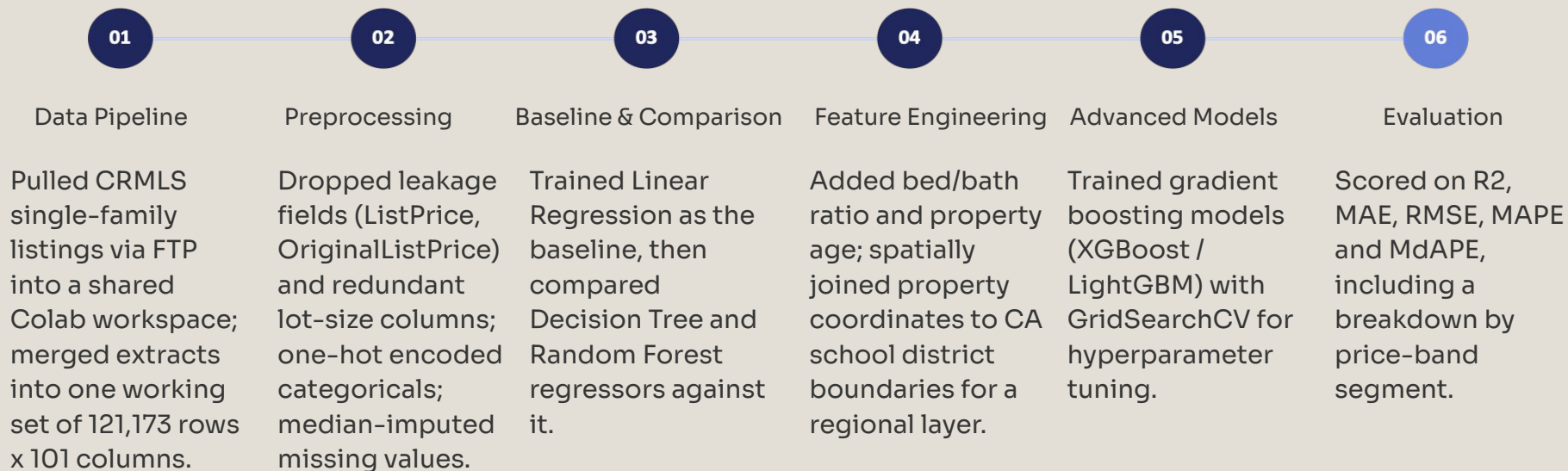
We simulated a real world market by testing the model on consecutive, unseen future periods

	Test_Window	Train_Size	Test_Size	R2	RMSE	MAE	MAPE (%)	MdAPE (%)
0	Window 1	278468	2320	0.855655	382239.608434	239834.460062	19.659389	14.473937
1	Window 2	280788	2320	0.848447	349539.058201	212496.708055	19.276760	13.331453
2	Window 3	283108	2320	0.850488	325934.508639	200549.063076	20.414496	13.727254
3	Window 4	285428	2320	0.814917	387562.872777	219876.019841	23.742771	13.998537
4	Window 5	287748	2323	0.797829	515994.440955	290917.774220	31.125136	17.827974



- Accuracy holds steady ($R^2 \approx 0.85$) across the immediate future market cycles (next 3 windows).
- Model is anchored on the data from the first 3 windows which included hundreds of thousands of rows, which caused it to miss new price trends and market shifts.
- Windows 4 and 5 had the most data but had the lowest performance. This is due to market shifts such as different interest rates, buy behavior, etc

Iteration Timeline



What Worked Best

01

Leakage-Safe Feature Set

Dropping ListPrice and OriginalListPrice, plus redundant lot-size columns, kept the model from cheating off features that echo the answer -- giving evaluation results the team could trust.

02

Gradient Boosting Upgrade

Moving from the Linear Regression / Random Forest baselines to a tuned XGBoost model produced the strongest overall fit: $R^2 = 0.898$, with MAPE of 13.8% and MdAPE of 9.1%.

03

Consistent Eval Pipeline

Reusing identical feature sets, encoding, and a fixed random_state across every model notebook made each iteration's results directly comparable to the last.

What Didn't Work

Weak Performance in Lower Price Bands

The overall model looks strong ($R^2 = 0.937$, MAPE 11.7%), but three of the five price quartiles (Q2, Q3, Q4) score a negative R^2 -- worse than guessing the segment average. This isn't a sign the model fails on those homes: their MAPE is actually better than the overall average. Once a band is narrow, prices vary so little that R^2 becomes an unreliable metric there -- MAPE and MdAPE tell the truer story.

The real accuracy gap is at the price extremes: Q1 (under \$575K) and Q5 (over \$1.67M) both carry the highest MAPE, around 14-15% -- cheap homes because a fixed dollar error is a bigger percentage, expensive homes because prices swing more.

XGBoost (Tuned) by Price Band

Segment	R^2	MAPE
Q1 (\$50K-575K)	0.53	14.3%
Q2 (\$575K-800K)	-0.65	8.4%
Q3 (\$800K-1.1M)	-1.36	10.0%
Q4 (\$1.1M-1.67M)	-0.54	10.9%
Q5 (\$1.67M-30M)	0.76	14.7%
Overall	0.937	11.7%

Duplicate Models

- Independently built XGBoost models on overlapping feature sets.
- The key difference was the train/test split: Nate's version used a time-based split (trained on Dec 2025–May 2026, tested on June 2026), matching how the model will actually be used in production. The alternative version used a random split, which doesn't test the model's ability to predict future months.
- Nate's version is presented here in detail: $R^2 = 0.937$ overall, MAPE = 11.7%, MdAPE = 7.8%.
- The team adopted Nate's time-based pipeline as the one shared model, rather than presenting both versions side by side.

Closing Summary

- The tuned XGBoost model reached $R^2 = 0.937$ overall (MAPE 11.7%, MdAPE 7.8%) on a time-based split — the strongest and most production-realistic result of the iteration cycle.
- Segment analysis showed R^2 becomes unreliable inside narrow price bands; MAPE points to the real gap, with the cheapest and most expensive homes hardest to price accurately.
- Adopting Nate's time-based pipeline let the team present one stronger, production-realistic model instead of competing versions.



Thank you!